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RESEARCH ARTICLE

Vertically oriented roots affect soil water infiltration by decomposition and altering soil structure: An examination based on CT technology and innovation in soil infiltration research methods

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Abstract
Aims This study aims to investigate how the dynamic process of root decomposition influences soil water infiltration by altering soil structure in forest ecosystems, addressing a critical knowledge gap in understanding soil hydrology under changing environmental conditions.

Methods In addition to measuring moisture infiltration, we used soil structural variables from computed tomography (CT) scans and tested whether CT-based soil structural variables correlated with soil moisture infiltration. Correlations between moisture infiltration and soil structure were analyzed by Pearson correlation. Direct and indirect effects of root decomposition time and soil structure on water infiltration were analyzed using structural equation modeling (SEM).

Results (1) root decomposition significantly affects soil water infiltration; (2) root decomposition changes soil stability infiltration rate by directly or indirectly affecting soil structure.

Conclusions This study shows root decomposition in *Cunninghamia lanceolata* and *Pinus taeda* plantations significantly improves soil water infiltration. Initial infiltration rates increased by 211% (*Cunninghamia*) and 445% (*Pinus*) over 360 days, driven by enhanced pore volume from organic matter release. Maximum stable infiltration and total water storage (103% and 143% increases) peaked at 240 days due to improved pore connectivity. A novel paraffin coating method combined with CT scanning quantified these changes non-destructively. Findings reveal root decomposition's critical role in enhancing soil hydrological function, offering insights for sustainable forest management. Future research should examine root traits (e.g., length density) and pore dynamics to refine mechanisms.

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Keywords Root decomposition · Soil structure · Paraffin coating method · Soil water infiltration · SEM

Introduction

Soil water is a key limiting factor for the sustainable development of plantation forests (Cui et al. 2019; Khan et al. 2009). Large-scale afforestation can intensify soil water depletion, which can cause soil desiccation or even the soil layers to dry in areas of limited precipitation (Jia et al. 2017a, b). Infiltration of surface water is extremely important to replenish soil water and meet the water demand for plant growth (Zhu et al. 2020). Rainfall can rapidly replenish groundwater through infiltration, thus achieving effective replenishment of forest soil water, which is an important part of the forest hydrological cycle (Liu et al. 2019). The infiltration capacity of forest soils is influenced by several factors. In previous studies, soil infiltration was found to be influenced by various soil properties and vegetation factors (Leung et al. 2015) such as soil bulk density (Fischer et al. 2014), soil organic matter content (Alaoui 2015), and roots (Jiang et al. 2018), which all affect soil structure. The effect of soil capacity on infiltration is governed by the size, number, and distribution of soil pores (Yimer et al. 2008), and infiltration capacity always decreases with increasing capacity (Hu et al. 2009; Liu et al. 2018). The effect of vegetation on infiltration is divided into two parts: above-ground and below-ground. Higher canopy density and aboveground biomass enhance soil water infiltration by reducing raindrop impact and surface exposure, thereby preserving soil structure. (Yimer et al. 2008). Reduced surface exposure, facilitated by canopy cover, minimizes raindrop impact, preventing soil compaction and crusting, which preserves soil pore structure and maintains infiltration capacity (Zhang and Xie 2019). The below-ground part of the plant changes soil properties through the roots and thus affects soil infiltration (Wang et al. 2019). Although a large number of studies have found that plant roots have a facilitating effect on soil infiltration, which increases with root length density, root surface area, and root diameter (Hao et al. 2020; Huang et al. 2019). At the same time, the roots can promote the increase of soil organic matter content, form soil aggregates, and thus increase soil porosity (Liu et al. 2020). However, root decomposition, which is an important component of root activity, has received little attention (Aulen et al. 2012; Kohout et al. 2021), and the effect of root decomposition on soil water infiltration is even less understood, which may lead to critical knowledge gaps in understanding the effect of roots on soil water infiltration.

Soil water infiltration is critical for sustainable forest ecosystems, as it facilitates groundwater recharge, reduces surface runoff, and mitigates flood risk at the landscape scale (Zhu et al. 2020; Neris et al. 2012). Effective infiltration replenishes soil water reserves, supporting plant growth and maintaining hydrological cycles in plantations (Liu et al. 2019). Conversely, poor infiltration can exacerbate soil erosion and flooding, particularly in areas with high rainfall intensity (Costantini et al. 2016). Vegetation, particularly plant roots, influences infiltration by altering soil structure and porosity (Jiang et al. 2018), yet the dynamic process of root decomposition and its impact on infiltration remains underexplored, limiting our ability to predict and manage forest hydrological processes.

Plant roots continuously change the morphological structure and the spatial distribution of their channels as they grow deeper into the soil (Cui et al. 2019), which determines the development of preferential flow channels (Jiang et al. 2018; Wu et al. 2017) and affect soil water infiltration (Fan et al. 2017). Root characteristics, such as root diameter, also determine the rate of root decomposition (He et al. 2019), which is an important source of soil organic matter (Six et al. 2004; Tang et al. 2024). As an important component of soil water-stable aggregates, soil organic matter significantly affects both soil porosity and water retention (Benegas et al. 2014) and has an important indirect role in soil infiltration. Roots increase soil organic matter and promote aggregate formation, enhancing soil porosity and water infiltration (Guo et al. 2019). Root channels formed by root decomposition can effectively improve soil infiltration properties (Huang et al. 2017) and accelerate horizontal and vertical water movement (Mitchell and Mitchell 1995; Tang 2024). As a result, deep soil water deficiency can be effectively supplemented by root decomposition (Weiler and Naef 2003), providing better water conditions for plant growth. Nevertheless, previous studies have seldom taken into account the effects of root decomposition processes on soil infiltration. Root decomposition, a key component of root dynamics, has received little attention, and its effects on soil water infiltration are poorly understood, highlighting the need for studies to inform sustainable forest management. Hence, it holds great significance to thoroughly examine the impact of plant roots, specifically root decomposition, on soil infiltration within forest ecosystems. Current research methods pose challenges in isolating root decomposition as a distinct factor, particularly when in situ soil core pretreatment is required, which limits our understanding of its effects on soil infiltration.

While various methods exist to measure soil infiltration and hydraulic conductivity (e.g., tension infiltrometers, rainfall simulators), this study focuses on double-ring and single-ring infiltration tests to evaluate their limitations in studying root decomposition effects on soil water infiltration (Wu et al. 2021; Zhang et al. 2017). The double-ring infiltration test, conducted in situ, cannot accommodate laboratory pretreatments (e.g., controlled root decomposition) or detailed soil structure analyses without disrupting the soil profile (Wu et al. 2021). After completing the double-ring infiltration experiment, a profile is excavated to obtain soil structure information. Profile excavation after double-ring infiltration may alter soil structure due to water movement, and root regrowth from adjacent plants may confound root decomposition studies, introducing measurement errors. First, the area where the above-ground parts of plants are removed is not necessarily the only area where root decomposition occurs; other plant roots in the soil may grow and occupy the original root space, which causes errors in root decomposition studies. Secondly, profile excavation was conducted after the completion of double-ring infiltration to obtain soil structural characteristics. However, the impact of water on the soil during the double- ring infiltration process will inevitably cause changes in soil structure, which will lead to errors in the results. Finally, excavation of profiles in the field is time-consuming and labor-intensive, and it is difficult to achieve a large number of replications of the experiment. Another method that is more commonly used is the laboratory single-ring infiltration test, which also suffers from the same problem as the double- ring infiltration method in that it is difficult to perform other treatments on the study object (Di Prima et al. 2019; Zhang et al. 2016). Meanwhile, the single-ring infiltration method is more demanding for collecting ring knife samples, and water often flows out from between the soil and the inner wall of the ring knife during infiltration experiments, which affects the reliability of the results. To overcome limitations of existing infiltration methods, we developed a novel laboratory single-ring infiltration test using paraffin-coated soil cores, detailed in Section "Permeability measurement", to study root decomposition effects on soil water infiltration.

To examine the alterations in forest soil water infiltration throughout root decomposition and gain insight into the impact of root decomposition-induced soil structure changes on water infiltration, we conducted a 12-month field experiment in *Cunninghamia lanceolata* and *Pinus taeda* plantations. We utilized an improved in situ soil core method for consistent root decomposition studies (Li et al. 2022) and a novel paraffin coating method for laboratory infiltration-test on intact soil cores (Section "Permeability measurement") (Bi et al. 2014; Di Prima et al. 2022, 2019) to examine root decomposition effects on soil water infiltration. This study aims to quantify the effects of root decomposition on soil water infiltration and soil structure in *Cunninghamia lanceolata* and *Pinus taeda* plantations using computed tomography (CT) and a novel paraffin-coated single-ring infiltration method. We hypothesize: (1) root decomposition increases initial, average, and stable infiltration rates; (2) root decomposition alters soil structure, particularly pore volume and connectivity; (3) changes in soil structure due to root decomposition enhance soil infiltration capacity; (4) root decomposition increases soil water storage by improving pore connectivity and water retention.

Materials and methods

Study site

Our study site was located at Xiashu Forest Farm, Jurong City, Jiangsu Province, China (119°22 ′46"E, 32°12′ 57"N), at an elevation of 100 m asl (Fig. 1). According to the World Soil Database (WHSD), the soils in the area were mainly leached soils with coarse topsoil texture, slightly acidic soils, high sand content, and low organic carbon content, with a soil reference depth of 1.0 m. The natural forest in the XiaShu Forest Farm was dominated by secondary oak forest.

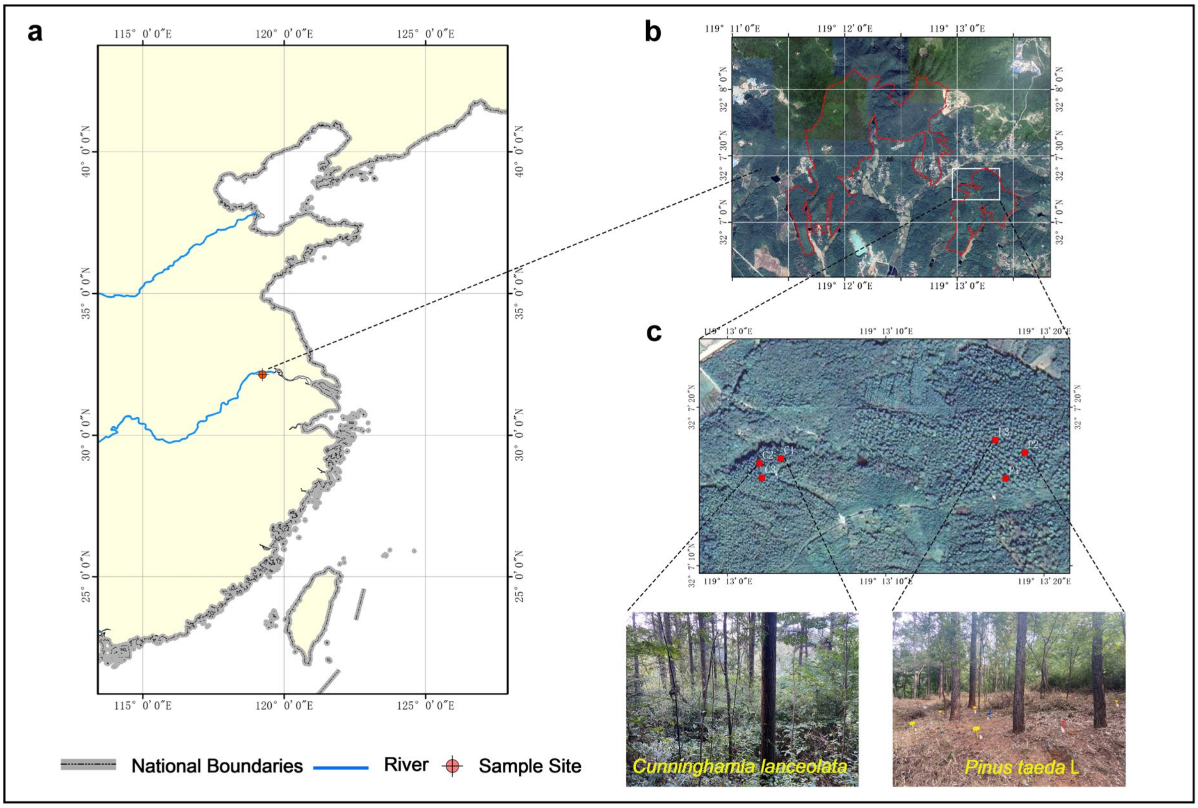


Fig. 1 Location map: **a** thumbnail of a map of China; **b** Location map of Xia Shu Forestry Field; **c** The sample site is located in the *Cunninghamia lanceolata* and *Pinus taedaL* plantation in the old mill area of Xia Shu Forestry

The region has a north subtropical monsoon climate. The average annual temperature was 15.2 °C. The lowest monthly average temperature was −0.8 °C in January. The highest average temperature was 31.6 °C in July. The average frost-free period was 233 days. The seasonal fluctuation of rainfall was large. The average annual precipitation was 1055.6 mm, and the average annual relative air humidity was as high as 79%. The total rainfall during the experimental period from October 2020 to October 2021 was 1167.4 mm, of which May–August precipitation accounts for more than 60% of the total precipitation.

Experimental design

The experiment was conducted in *Cunninghamia lanceolata* and *Pinus taeda* plantations at Xiashu Forest Farm. Three 20 m × 20 m plots were randomly established in each plantation type, ensuring uniform stand conditions. Within each plot, five sample strips were set up, located at least 1 m from the nearest tree to minimize variation in root biomass (Dornbush et al. 2002; Li et al. 2022). Soil cores (10 cm diameter, 60 cm depth) were collected using a custom stainless-steel corer (10 cm inner diameter, 100 cm length) designed to extract intact 60 cm deep cores with minimal disturbance. Four cores were collected per strip, yielding 20 cores per plot and 60 cores per plantation type (Table 1).

Soil cores were placed in 200-mesh nylon bags to allow soil contact while preventing root intrusion, returned to their original locations, and buried to reflect natural decomposition conditions (e.g., ambient temperature, rainfall, and microbial activity). Cores were retrieved in October 2020 (0 days), February 2021 (120 days), June 2021 (240 days), and October 2021 (360 days) to capture different stages of root decomposition. One core per strip was retrieved at each time point, resulting in five cores per plot per

Table 1 Summary of experimental design and measurements

Plantation Type	Plots	Sample Strips per Plot	Cores per Strip	Total Cores per Plot	Decomposition Time Points (days)	Measurements Performed
Cunninghamia lanceolata	3	5	4	60	0, 120, 240, 360	CT scan, infiltration, soil structure analysis
Pinus taeda	3	5	4	60	0, 120, 240, 360	CT scan, infiltration, soil structure analysis

time point. To control for soil moisture variability, the water content of cores was measured at day 0 and before each subsequent retrieval, ensuring differences were within ± 5% of the initial measurement. In each sampling session, the in situ soil cores that were buried in nylon mesh bags were retrieved, protected in 10 cm × 60 cm polyvinyl chloride (PVC) pipes, and scanned using CT within 12 h.

Permeability measurement

To realize the permeability measurement of original soil cores with different times of root decomposition, we designed and invented a new permeability measurement device and method about the single ring infiltration method (Paraffin coating method). The method has been examined and licensed by the State Intellectual Property Office of China (Patent No. ZL 2021 1 0658926. X). The specific steps of the method are as follows:

Step 1: Carefully remove the original soil core from the protective PVC case and nylon mesh bag after completing the CT scan, and remove the debris and soil particles attached to the surface of the original core (Fig. 2a).
Step 2: Carefully place the original soil core into the fixture (Fig. 2b).
Step 3: Lift the original soil core with its fixture using a fixed tripod and pulley set and repeatedly submerge it in the paraffin solution (Fig. 2c). (Note: The paraffin granules were melted at 80 °C using an electric heater and cooled to 50–55 °C for 1 h to achieve a viscous, semi-solid consistency, preventing pore clogging during coating. The completely melted paraffin liquid will enter the original core through the soil pores in the sidewalls of the original core, thus blocking the soil pores and affecting the permeability determination.)
Step 4: Make sure that the paraffin last wrapped in the original core is solidified and tightly adhered to the original core before each intrusion of the melted paraffin liquid into the original core (Fig. 2d). (Note: leave the top of the original core in place each time the original core is inserted into the paraffin solution, do not submerge the entire original core in the paraffin liquid)
Step 5: Repeat the above steps 5 times to ensure that the paraffin shell is strong enough and fits tightly. Carefully remove the paraffin-wrapped original core from the fixture (Fig. 2e).
Step 6: Wrap the top part of the original soil core in contact with the soil using waterproof raw tape (to prevent infiltration rate measurement as water leaks through the joints). Attach a pre-prepared clear plastic ring (10 cm in height) with a diameter slightly larger than the diameter of the core to the top of the core. The joints are tightly wrapped with waterproof raw tape and transparent tape, and the interface is waterproofed with petroleum jelly inside the transparent plastic ring to prevent water leakage during the infiltration test. After completing the above work, the cores were cut and only 60 cm lengths were retained (the cores were drilled to an additional depth to ensure that the core lengths met the test design requirements). Petroleum jelly was applied only to the inner surface of the plastic ring-core interface to seal gaps, preventing water leakage. Post-experiment dissection confirmed no jelly intrusion into soil pores. The cores were placed in a custom-made Plexiglas cylinder (12 cm diameter, 60 cm height) with a perforated base to collect outflow. A Mariotte's bottle maintained a constant 3 cm water head during infiltration tests (Fig. 2f).
Step 7: Fill the clear plastic ring at the top of the original soil core with pure water was added to maintain a 3 cm head, controlled by a Mariotte's bottle, ensuring consistent infiltration conditions.

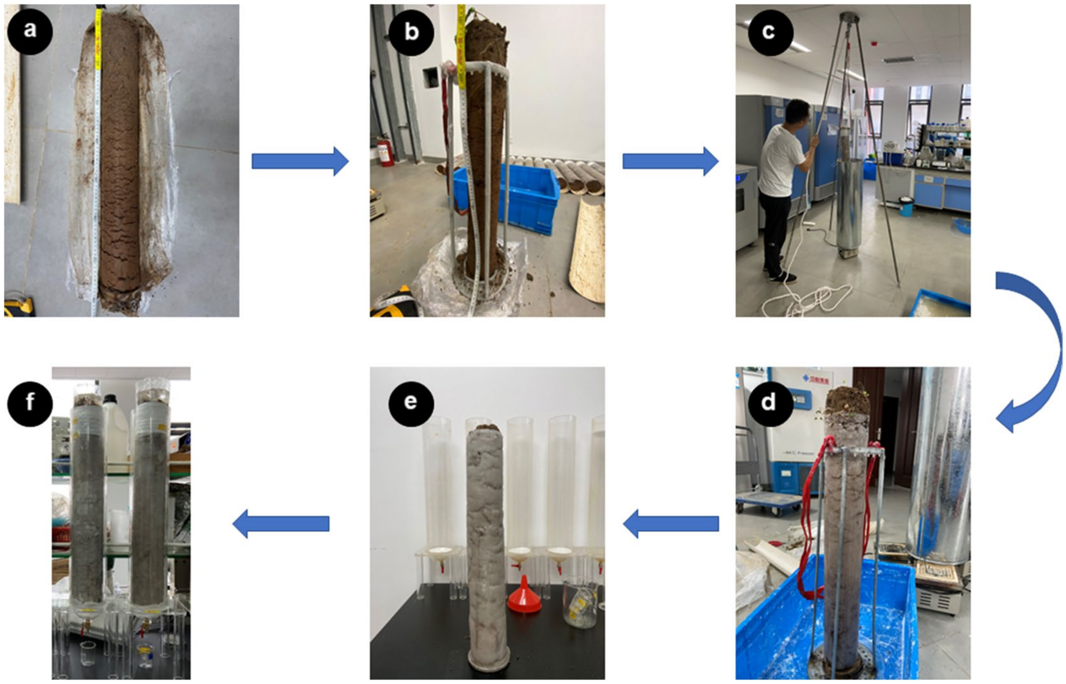


Fig. 2 Preparation of soil infiltration: **a** Remove in-situ soil core from PVC packaging; **b** Carefully place the original soil core into the fixture; **c** Repeated immersion of the in-situ soil core into the melted paraffin liquid using a lifting device; **d** Allow the paraffin wax wrapped on the surface of the in-situ soil core to solidify before the next intrusion into the paraffin liquid; **e** Repeat the intrusion of paraffin liquid 5 times and then remove the in-situ soil core and leave it to solidify completely; **f** The bottom excess is cut off and the top of the in-situ soil core is wrapped with waterproof tape and then placed in the infiltration device for infiltration testing

The total amount of pure water injected into the clear plastic ring for a fixed time was recorded separately during the experiment. The recording interval was 5 min for each of the first four intervals and 20 min for each of the subsequent intervals (5 min, 10 min, 15 min, 20 min, 40 min, 60 min, 80 min, and so on).

Step 8: A measuring cup was placed at the device's base to collect outflow, and the outflow initiation time (minutes) was recorded to quantify the speed of water movement through the soil core, indicating pore connectivity.

Step 9: Terminate the experiment when the same amount of pure water is injected into the transparent plastic ring 3 consecutive times per unit time.

Step 10: Cores were dissected post-experiment, and visual inspection identified paraffin intrusion as white, waxy material within soil pores; affected cores were discarded. Approximately 5% of cores (3 out of 60 per plantation type) were discarded due to paraffin intrusion, detected during post-experiment dissection.

Permeability-related variables were calculated (Bi et al. 2014; Wu et al. 2016) as follows:

Initial Infiltration Rate (IIR): $IIR = \Delta I_5/5$ (mm/min), where ΔI_5 is the total infiltration (mm) in the first 5 min.

Stable Infiltration Rate (SIR): $SIR = \Delta I_{20}/20$ (mm/min), where ΔI_{20} is the total infiltration (mm) during a 20-min period when infiltration stabilizes (> 20 min).

Average Infiltration Rate (AIR): $AIR = \Delta I/\Delta t$ (mm/min), where ΔI is the total infiltration (mm) at stable infiltration, and Δt is the time (min) to reach stability.

Total Soil Water Storage (TSW): $TSW = \Delta I — \Delta RP$ (mm), where ΔRP is the total runoff (mm) (Runoff (ΔRP) was quantified as the volume of water collected in a measuring cup at the infiltration device's base, converted to depth (mm) based on the core's cross-sectional area.).

In this study, runoff is defined as the water production occurring at the bottom of the infiltration device. When the original soil core in the laboratory is completely saturated, water production emerges below the core due to the soil's inability to absorb more water. In the field, this excess water that cannot be absorbed manifests as surface runoff. Cumulative infiltration rate was calculated as the total volume of water infiltrated over time (mm), determined by summing infiltration measurements at each recording interval.

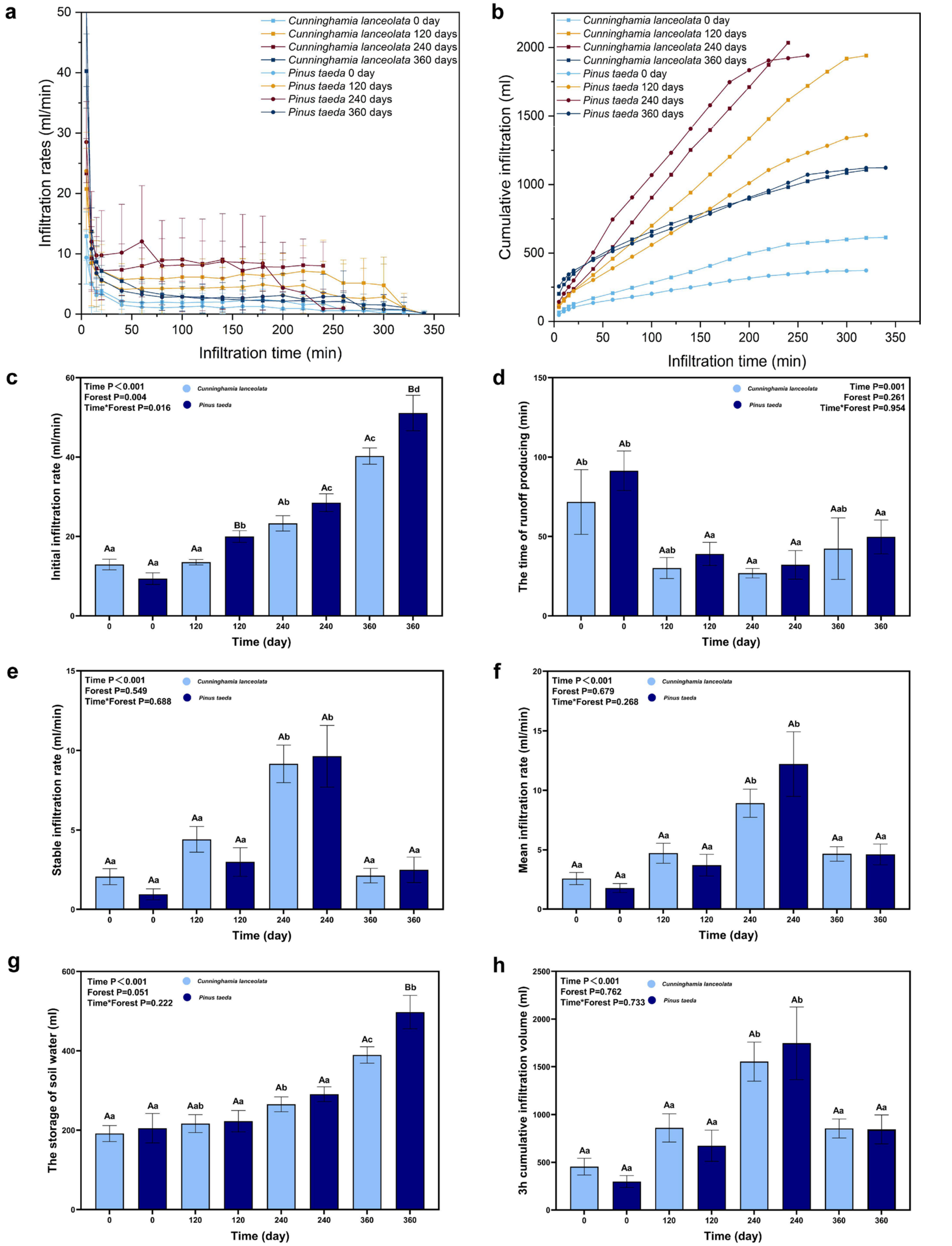
The infiltration variables were selected to capture different aspects of soil water dynamics: Initial Infiltration Rate (IIR) reflects surface pore accessibility and rapid water entry; Stable Infiltration Rate (SIR) indicates long-term infiltration capacity under saturated conditions; Average Infiltration Rate (AIR) provides an overall measure of infiltration efficiency; and Total Soil Water Storage (TSW) quantifies the soil's capacity to retain water, critical for plant growth and groundwater recharge (Bi et al. 2014; Wu et al. 2016).

Soil structures

The intact soil cores were scanned using a United Imaging uCT510 medical scanner (United Imaging Intelligence, Shanghai, China). Scanning was done at 1 mm intervals using 120 kV and 335 mA current. The scanned images were reconstructed using Avizo 2019.1 (Thermo Fisher Scientific Avizo, Waltham, USA) and image correction was performed using VG Studio Max 2022.1 (Volume Graphics GmbH, Germany). The images were thresholded for segmentation using Avizo's intensity histogram, and the soil pore was reconstructed and then analyzed using Avizo's computational module (Katuwal et al. 2015) (Table 2).

Table 2 The characteristics of the soil structures

Parameters	Explanatory notes
The volume fraction of soil	the volume of soil/volume of undisturbed soil core
Length	Maximum of the Feret Diameters Feret diameter is defined as the distance between two tangent planes of a particle in a given direction
Breadth	Largest distance between two parallel lines touching the object without intersecting it and lying in a plane orthogonal to the maximum Feret diameter
Width	Minimum of the Feret Diameters
Thickness	The largest segment that touches the object by its endpoints and lying in a plane orthogonal to the maximum Feret diameter and orthogonal to the breadth diameter
OrientationPhi	Phi orientation of the particle in degrees [0, +90], computed with the inertia moments. It defines with OrientationTheta the eigenvector of the largest eigenvalue of the covariance matrix
OrientationTheta	Theta orientation of the particle in degrees [−180,180], computed with the inertia moments. It defines with OrientationPhi the eigenvector of the largest eigenvalue of the covariance matrix
Shape	The shape factor is defined as $\text{shape} = \text{Area}^3/(36 \times \pi \times \text{Volume}^2)$
Tortuosity	Tortuosity is defined as the ratio between the length of the path and the distance between its ends along the z-axis. In our case, the distance between the ends of the curve is given by the number of planes along the z-axis
Fractal dimension	A fractal dimension is a number greater than 2 and strictly lower than 3. The result is 2 in the case of standard geometric surfaces. Applied to 3D images the fractal dimension is quite an effective indicator to measure and compare the roughness of a surface. It is also a good indicator to evaluate how the curve fills the space. The less smooth the surface is, the bigger the fractal dimension. It can also be interpreted as a quantification of how complex the surface is and how it fills the space
Flatness	The ratio of the smallest to the medium eigenvalue of the covariance matrix
EqDiameter	The equivalent diameter is the diameter of the sphere of the same volume
VoxelFaceArea	This value is the sum of voxel surfaces that are on the outside of each connected component
Euler	Connectedness indicator. It is an indicator of the connectivity of a 3D complex structure


a
Cunninghamia lanceolata 0 day
Cunninghamia lanceolata 120 days
Cunninghamia lanceolata 240 days
Cunninghamia lanceolata 360 days
Pinus taeda 0 day
Pinus taeda 120 days
Pinus taeda 240 days
Pinus taeda 360 days
Infiltration rates (ml/min)
Infiltration time (min)
b
Cumulative infiltration (ml)
Infiltration time (min)
c
Time P＜0.001
Forest P=0.004
Time*Forest P=0.016
Cunninghamia lanceolata
Pinus taeda
Initial infiltration rate (ml/min)
Time (day)
d
Time P=0.001
Forest P=0.261
Time*Forest P=0.954
The time of runoff producing (min)
Time (day)
e
Time P＜0.001
Forest P=0.549
Time*Forest P=0.688
Stable infiltration rate (ml/min)
Time (day)
f
Time P＜0.001
Forest P=0.679
Time*Forest P=0.268
Mean infiltration rate (ml/min)
Time (day)
g
Time P＜0.001
Forest P=0.051
Time*Forest P=0.222
The storage of soil water (ml)
Time (day)
h
Time P＜0.001
Forest P=0.762
Time*Forest P=0.733
3h cumulative infiltration volume (ml)
Time (day)

◂**Fig. 3** Soil infiltration: **a** Infiltration rate curves; **b** Cumulative infiltration curves; **c** Initial infiltration rate; **d** The time of runoff production; **e** Stable infiltration rate; **f** Mean infiltration rate; **g** The storage of soil water; **h** 3 h cumulative infiltration volume (Note: upper case letters indicate significant differences between different stand type ($p < =0.05$); lower case letters indicate significant differences between different time ($p < =0.05$); the light blue columns indicate *Cunninghamia lanceolata* plantations; dark blue columns indicate *Pinus taedaL.* Plantations)

Statistical analysis

Each stand type had three replicate sample plots, and each sample plot included five replicates for the same root decomposition time. During the one-year experiment and infiltration test, data unusable for the infiltration experiment were excluded due to the damage of undisturbed soil cores caused by external factors (such as human destruction, wild animal activities, transportation accidents, etc.), which led to paraffin intrusion into the soil cores. (As the experimental method was pioneered by the author without specific protocols, operational errors inevitably occurred during the experiment, causing paraffin liquid to seep into the soil cores through lateral pores, rendering them unusable.) Three replicates per decomposition time per plot were retained for analysis (9 per plantation type) after excluding cores with paraffin intrusion or physical damage, determined by visual inspection during post-experiment dissection. CT data were reduced to match the retained cores.

All statistical analyses were performed using SPSS 26.0 (SPSS Inc., Chicago, IL, USA). To detect differences in infiltration rate, and soil structure between different time-to-root decomposition and different stand types, we first analyzed the interaction between time-to-root decomposition and stand types using a two-way analysis of variance. With one exception, we detected a significant interaction between the time to root decomposition and stand type. Therefore, we conducted and reported one-way ANOVA for stand type separately for each time-to-root decomposition and time-to-root decomposition for each stand type. If ANOVA results were significant ($p < 0.05$), the least significant difference (LSD) test was used to determine significant differences between time to root decomposition and stand type ($p < 0.05$).

Principal component analysis (PCA) was performed on soil structure variables to reduce dimensionality and identify key parameters influencing infiltration (Liu et al. 2022). Based on the results of the Kaiser–Meyer–Olkin test and Bartlett's spherical test, the data were confirmed to be suitable for PCA. The most critical parameters (Taylor and Tibshirani 2015) identified from the principal component analysis (PC1 and PC2) were used to determine the soil structure parameter scores (Fig. 4):

$$Y = (39.2Y_{PC1} + 30.6Y_{PC2})/69.8$$

where Y is the score for the soil pore parameters. Y_{PC1} and Y_{PC2} are the scores for PC1 and PC2, respectively. These tests were performed in the Origin 2021 software.

We used Pearson's correlation coefficient to quantify the strength and significance of the relationship between soil structure, time-to-root decomposition, stand type, and infiltration rate. Structural equation modeling (SEM) analysis (Amos, 24.0 Chicago, IL, USA) was used to estimate the direct and indirect effects of soil structure and time-to-root decomposition on infiltration rates. Several metrics ($\chi 2$, p-value, GFI, and RMSEM) were used to test the goodness of fit of the model: p-value > 0.05, GFI > 0.9, and RMSEM < 0.08. Graphs were plotted using Origin (Origin, version 2021, Northampton, MA, USA) and GraphPad Prism 9 (GraphPad Software, Inc., La Jolla, CA, USA).

Results

Soil infiltration rate at different root decomposition times

In general, the infiltration rate of field-collected soil cores at different root decomposition times showed a decreasing trend with increasing infiltration time during the infiltration experiment. Specifically, the infiltration started to decrease faster at the beginning (0—50 min) and leveled off at the end (50—300 min) (Fig. 3a). and the infiltration started to decrease faster at the beginning (0–50 min) and leveled off at the end (50–300 min) (Fig. 3a). The cumulative infiltration rate increased with the increase in infiltration time (Fig. 3b). There were no significant differences in infiltration rate variables between stand types except for the initial infiltration rate, which was

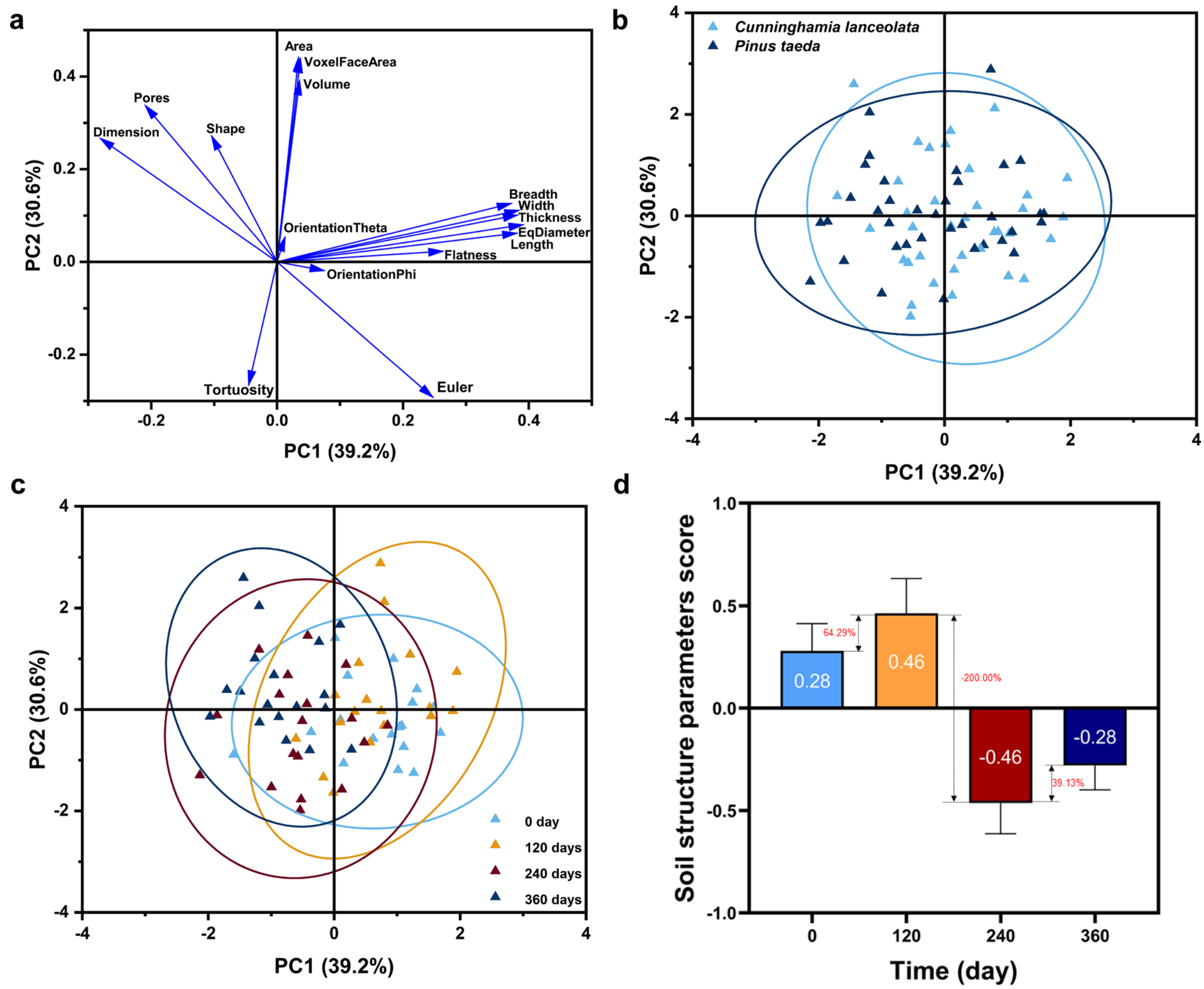


Fig. 4 Principal component analysis (PCA) of soil pore-related variables: **a** Variation in the first two PCA axes; **b** Variation in the first two PCA axes between the two stand types; **c** Variation in the first two PCA axes across four different times of decomposition; **d** PCA-based soil pore score at different time of decomposition

higher in *Pinus taeda* than in *Cunninghamia lanceolata* plantations at the decomposition times 120, 240, and 360 days (Fig. 3c). Initial infiltration rate and soil water storage increased significantly with increasing root decomposition time (Fig. 3c, g), with 211.34% (*Cunninghamia lanceolata*) and 445.02% (*Pinus taeda*) increase in initial infiltration rate and 103.36% (*Cunninghamia lanceolata*) and 142.51% (*Pinus taeda*) increase in soil water storage throughout the experimental cycle. The in-situ soil cores at the root decomposition time 0 day produced significantly higher runoff flow than in-situ soil cores of other times (Fig. 3d). Stable infiltration rate, average infiltration rate, and total 3 h infiltration increased with increasing root decomposition time and reached a maximum at 240 days of root decomposition, followed by a decline (Fig. 3e, f, h).

Changes in soil structural parameters altered water infiltration

The correlation results showed a significant correlation between the infiltration rate of in-situ soil cores and soil structure variables at different times of root decomposition (fig a). Soil structure parameter scores indicated increased pore volume fraction and connectivity in the early stages of root decomposition (64.29% increase at 120 days compared to 0 days), enhancing water movement, while lower scores were observed in the middle stages of root decomposition (−200% decrease at 240 days compared to 120 days). Finally, there was a rebound in the late stage of root decomposition (39.13% increase in the score at 360 days compared to 240 days) (Fig. 4).

Effect of soil structure on soil infiltration rate after root decomposition

First, the time of root decomposition was significantly correlated with all soil infiltration variables except stable infiltration rate. Only a significant negative correlation was found with the original soil core flow production time. Second, 62.5% of the soil structure variables were significantly correlated with root decomposition time and initial infiltration rate. Finally, the mean and stable infiltration rates were significantly and positively correlated with Dimension and shape, and significantly and negatively correlated with Width, Length, EqDiameter, and Thickness (Fig. 5). As root decomposition proceeded, the volume fraction of soil continued to increase, and all indicators of infiltration (Initial infiltration rate (Fig. 6a), Stable infiltration rate (Fig. 6c), Mean infiltration rate (Fig. 6d), The storage of soil water (Fig. 6e), 3 h cumulative infiltration volume (Fig. 6f)) increased with the increase in the volume fraction of soil, except for the shortening of the time of runoff producing (Fig. 6b).

Structural equation modeling (SEM) indicated that Shape (path coefficient = 0.224) had a direct positive effect on stable infiltration rate, while Thickness (path coefficient = −0.037) had a direct negative effect on stable infiltration rate. Root decomposition time (path coefficient = 0.235) had an indirect effect on stable infiltration rate by changing soil structure variables (Fig. 7).

Discussion

Root decomposition significantly affects soil water infiltration

Root decomposition significantly increased initial infiltration rate (IIR) by 211% and 445% for *Cunninghamia lanceolata* and *Pinus taeda*, respectively, over 360 days, driven by increased pore volume fraction. Pearson's correlations showed strong positive relationships between IIR and pore volume fraction ($r = 0.51$, $p < 0.001$) and shape ($r = 0.45$, $p < 0.001$), confirming that larger, interconnected pores facilitate rapid water entry. Previous studies also reported that root decomposition promoted the formation of soil pores (Guo et al. 2019) and increased soil infiltration and water storage capacity (Cui et al. 2019). As root decomposition is faster in upper soils (Hicks Pries et al. 2018), more pores are formed in upper soils, resulting in the increase of the initial infiltration rate. This increase in initial infiltration rate can achieve rapid replenishment of deep soil water. In the field, rainfall events can directly affect surface runoff and groundwater levels (Jiang et al. 2018). The 103% and 143% increases in soil water storage for *Cunninghamia lanceolata* and *Pinus taeda*, respectively, highlight root decomposition's role in enhancing water retention. These findings inform soil conservation strategies, as improved infiltration reduces runoff and flood risk (Neris et al. 2012). Future studies should examine root length density and pore size distributions to further elucidate these mechanisms(Costantini et al. 2016; Neris et al. 2012).

From our results, we found that the stable and average infiltration rates of the soil did not increase unrestrictedly with increasing root decomposition time, but peaked at a root decomposition time of 240 days. During the 240—day experimental period, it corresponds to the rainy season (June) of the subtropical monsoon climate, with an average daily temperature of 25.3℃. At this time, the activity of thermophilic bacteria reaches its peak, accelerating the degradation of root cellulose. The released polysaccharide substances act as cementing agents to maintain the pore structure (Wu et al. 2019, 2016). Compared with the 360—day period (autumn, with an average daily temperature of 18.2℃), anaerobic decomposition dominates, leading to the accumulation of organic acids, which prompts the expansion of clay minerals to fill the 100—200 μm pores, and the saturated hydraulic conductivity decreases. The 240—day period during the experiment is exactly at the inflection point of temperature change, resulting in the peak of infiltration during the 240—day period.. Humus in organic matter is a colloidal substance with strong adsorption capacity, which can cement soil particles (such as sand, silt, and clay) together to form water-stable aggregates. Meanwhile, as the main energy source for

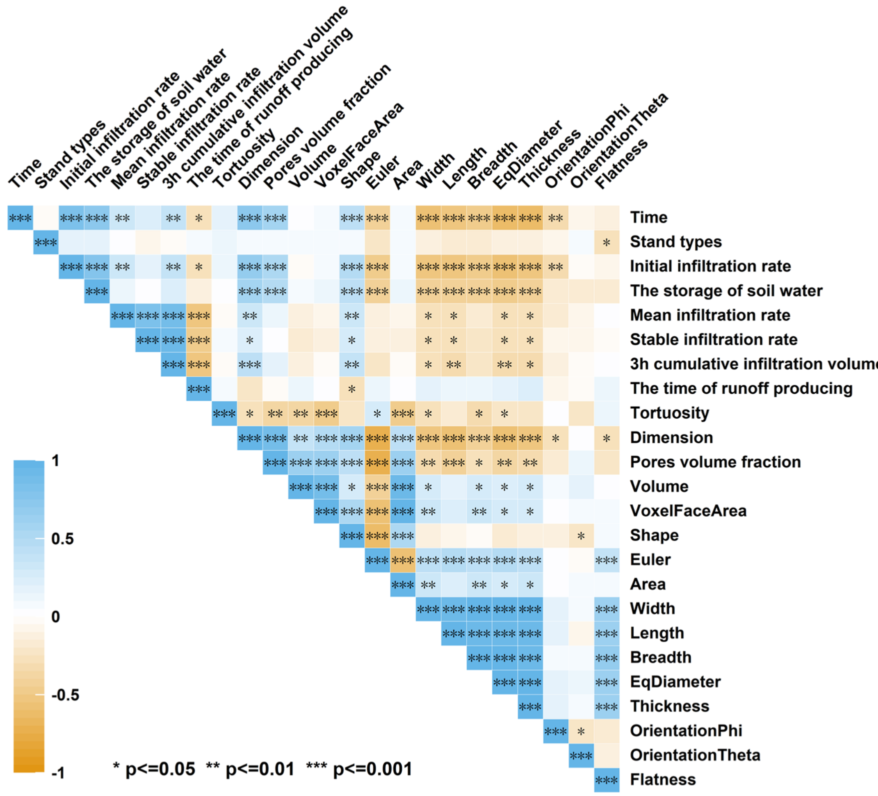


Fig. 5 Pearson correlation between time of decomposition, stand types, soil infiltration, soil structures (*** indicates significant correlation at $p<=0.001$, ** indicates significant correlation at $p<=0.01$, * indicates significant correlation at $p<=0.05$)

soil microorganisms, organic matter is decomposed by microbial activities to secrete viscous substances such as polysaccharides and uronic acids, which further promote aggregate formation. These aggregates fill the originally connected macropores, leading to the decline of saturated hydraulic conductivity in accordance with the prediction of the Kozeny-Carman equation. Increased water retention, driven by organic matter release and aggregate formation, reduces stable and average infiltration rates at 360 days by filling pores with water, limiting further infiltration (Cui et al. 2019; Liu et al. 2020). Therefore, when the root decomposition time reaches 360 days, the original soil core may be in a high water content environment for a long time, resulting in a significant thickening of the pore water film. This thickened water film will impede the passage of water, thus reducing the stable infiltration rate and the average infiltration rate compared to the previous stage. At the same time, the long—term high water content environment may accelerate the decomposition of soil organic matter and the formation of soil aggregates (as observed in the 360—day treatment group). The increase in water—stable aggregates will fill part of the pore space, especially reducing the proportion of macropores, thus reducing the saturated hydraulic conductivity of the soil (Rawls et al. 2004; Six et al. 2000). However, the initial infiltration rate is the infiltration rate 5 min before the start of infiltration, and this part of infiltration is more closely related to the upper soil. The water content of the upper soil is generally affected more by the external environment (Liu et al. 2021), which leads to the difference in infiltration rate variation.

Root decomposition changes soil stability infiltration rate by directly or indirectly affecting soil structure

Our study provides direct evidence that root decomposition leads to an increase in soil pore volume fraction and significantly increases soil infiltration rate (Beven and Germann 1982). From studies on the root decomposition of shrubs (Guo et al. 2019) and trees (Wu et al. 2021), it was found that root decomposition creates channels for preferential infiltration of available water (Mitchell et al. 1995). It is well known that

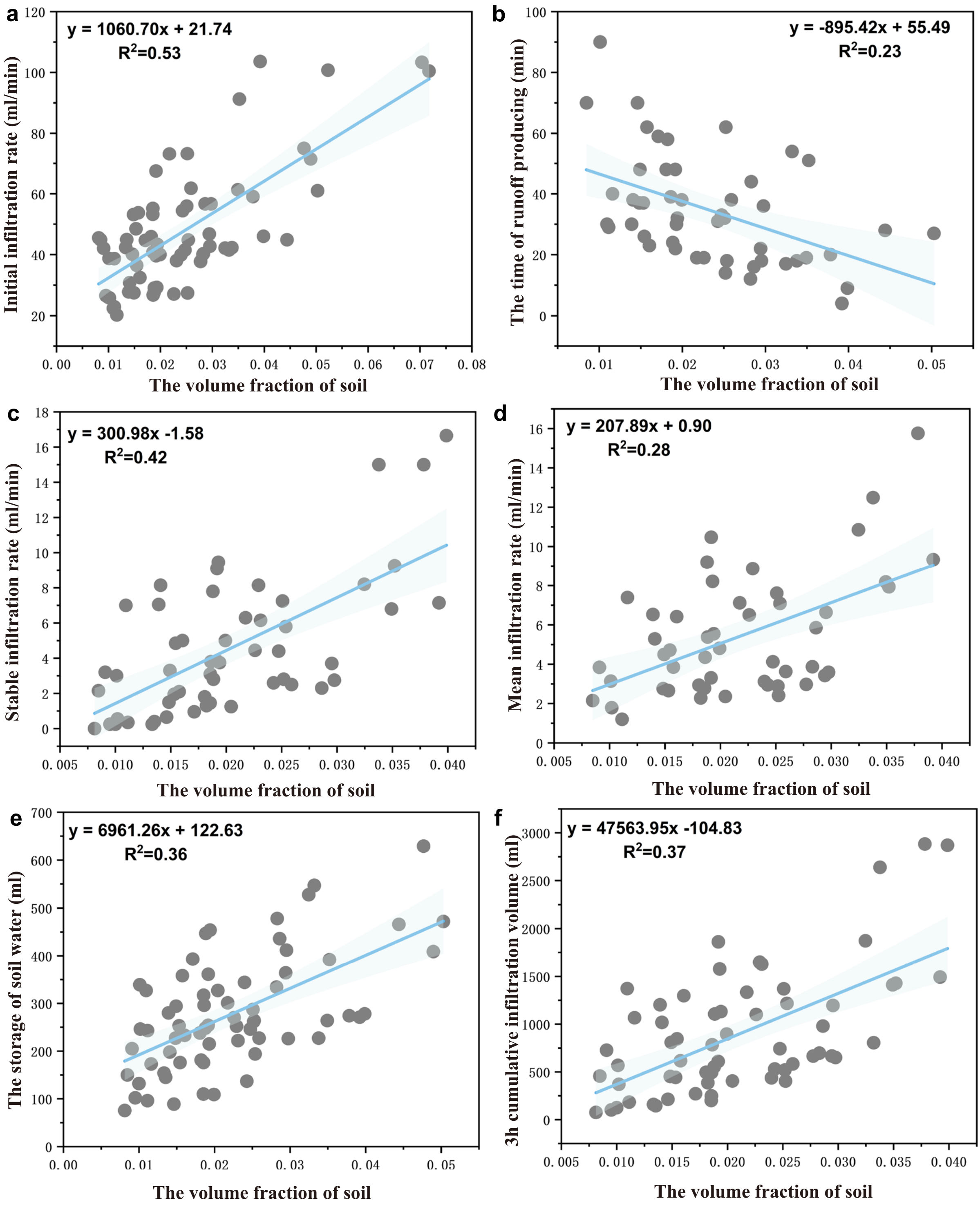


Fig. 6 Scatter plots showing the relationship between infiltration and the volume fraction of soil: **a** Initial infiltration rate (ml/min); **b** The time of runoff producing (min); **c** Stable infiltration rate (ml/min); **d** Mean infiltration rate (ml/min); **e** The storage of soil water (ml); **f** 3 h cumulative infiltration volume (ml)

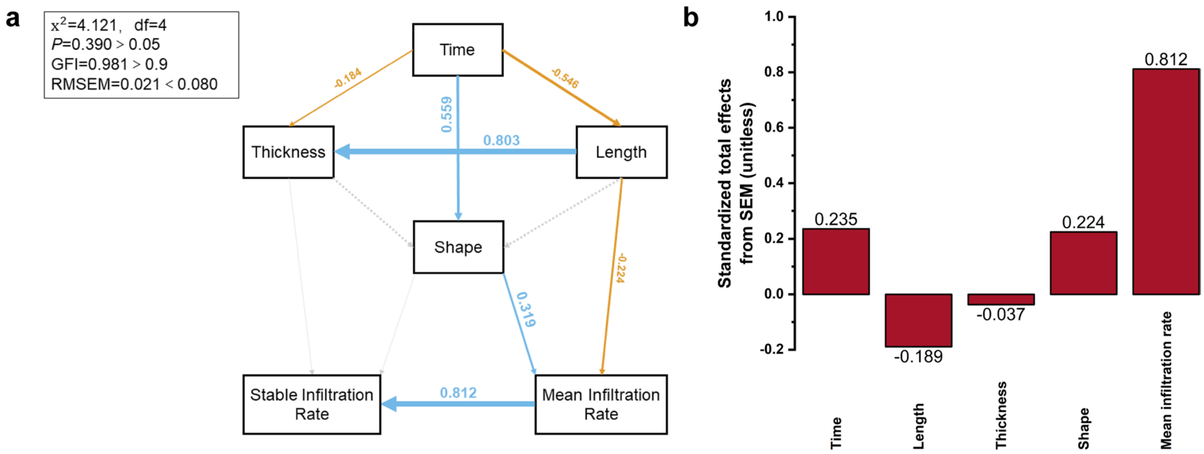


Fig. 7 **a** Structural equation model (SEM) analysis estimating the direct and indirect effects of CT scan data of the soil structures on stable infiltration rate. Boxes show variables included in the model. Test results of the goodness of model fit Chi-square (2) = 4.121, *p*-value = 0.390 > 0.05, goodness-of-fit index (GFI) = 0.981 > 0.9, root square mean error of approximation (RMSEA) = 0.021 < 0.08. Numbers on arrows are standardized path coefficients. The widths of the arrows represent the strength of the relationships. Blue arrows indicate positive relationships, and yellow arrows indicate negative relationships. Solid arrows indicate significance ($p < 0.05$) and dashed arrows represent non-significance ($p > 0.05$) (the p-value is calculated in terms of the nonnormalized path). **b** Standardized total effects (direct plus indirect effects) derived from the structural equation models depicted above

water movement in soil is mainly influenced by soil pores, and an increase in soil porosity will inevitably promote soil water infiltration. Root growth also changes soil structure (Johnson and Lehmann 2006), and soil pores provide more favorable conditions for root growth, which promotes pore connectivity. When the roots decompose, the space originally occupied by the roots is released, which in turn enhances soil porosity for water transport. However, we observed in our year-long study that the length and width of soil pores appeared to decrease to some extent as root decomposition progressed (Fig A2). We speculate that this may be due to the continuous release of fine soil pores after fine root decomposition, which pulls down the average size of soil pores. It could also be due to the root residues being transported and blocking some of the pores, resulting in a reduction in pore size. However, the total volume fraction of soil pores increased (fig A2a), and the reduction in soil pore length and width did not act as a barrier to soil water infiltration.

Stable infiltration rate (SIR) peaked at 240 days, with shape (path coefficient = 0.224) exerting a direct positive effect and thickness (path coefficient = −0.037) a negative effect, as shown by SEM. This suggests that complex, interconnected pores enhance long-term infiltration, while thicker pore walls impede flow. The decline in SIR at 360 days likely results from increased water retention due to organic matter release, which forms water-stable aggregates (Cui et al. 2019; Liu et al. 2020). The core mechanism of the 360—day SIR decline is the pore—structure transformation induced by organic matter. The organic matter released by root decomposition forms complexes with clay minerals, filling the soil pores. At the same time, water—stable aggregates compress the pore—connectivity network. The shape is a variable used to characterize the 3D structure of the soil pore (Li et al. 2019). Water-conducting pores in soil consist of many interconnected pores, forming a large network, resulting in a large shape. these water-conducting pores may be isolated water storage spaces under normal conditions and do not conduct water. When root decomposition is further intensified, more pores are created in the soil. The concomitant water infiltration generates sufficient water pressure leading to the expansion of the pores and promoting soil water infiltration (Li et al. 2018). Overall, both root growth and decomposition promote an increase in soil organic matter and soil porosity, which in turn changes soil structure. However, root decomposition has a more direct contribution to soil porosity than root growth.

Conclusion

Our study provides new and strong evidence for the influence of root decomposition processes on water infiltration in forest soils. Consistent with our initial hypotheses, our results indicated that the initial soil infiltration rate increased significantly, by 211% for *Cunninghamia lanceolata* and 445% for *Pinus taeda*, with increasing root decomposition time. Stable soil infiltration rate and total infiltration, influenced by the water retention capacity of the soil, reached a maximum at a root decomposition time of 240 days. The soil structure was improved by root decomposition, the pore volume fraction increased continuously, and the amount of water that could be stored in the soil increased continuously by 103% for *Cunninghamia lanceolata* and 143% for *Pinus taeda*. In addition, root decomposition promoted the increase of soil pore space, which effectively improved the infiltration and water retention capacity of the soil. These findings are revealing for understanding the influence of the root decomposition process on water infiltration and retention in forest soils. Our study also contributes to a better understanding of the forest soil environment and provides a theoretical basis for soil and water conservation in sustainable forest management. Limitations include the focus on two plantation types, which may not represent broader forest ecosystems, and the lack of root trait measurements (e.g., root length density). Root length density (RLD) likely enhances infiltration by increasing pore connectivity, but was not measured in this study due to the focus on decomposition. Future research should quantify RLD and other root traits (e.g., diameter, surface area) to elucidate their contributions to infiltration dynamics. Field conditions (e.g., rainfall variability) may also introduce variability. Future studies should investigate diverse species, quantify root traits, and classify pore size distributions to refine the mechanistic understanding of root decomposition's effects on infiltration.

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Data Availability The datasets generated during this study are fully available within the article and supplementary materials.

Declarations

Conflicts of interest The authors declare that they have no competing interests.

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